

The space $H_0^1(\Omega)$ is a Hilbert space with the H^1 norm $\|\cdot\|_{1,\Omega}$. Since $F \in H^{-1}(\Omega)$, it is a continuous linear functional on this space. The assumed boundedness and coercivity of a therefore allow the Lax–Milgram lemma to be applied, giving a unique $y \in H_0^1(\Omega)$ such that $a(y, v) = F(v)$ for every $v \in H_0^1(\Omega)$.

The finite-dimensional subspace V_h^0 , equipped with the inherited norm, is also a Hilbert space. The restrictions of a and F to V_h^0 remain bounded, and the restriction of a remains coercive with the same constant c . A second application of the Lax–Milgram lemma thus gives a unique $y_h \in V_h^0$ satisfying $a(y_h, v_h) = F(v_h)$ for every $v_h \in V_h^0$.

Set $e = y - y_h$. Subtracting the two variational equations for any $v_h \in V_h^0$ yields Galerkin orthogonality: $a(e, v_h) = 0$. Now fix an arbitrary $v_h \in V_h^0$. Because $y_h - v_h \in V_h^0$, orthogonality, coercivity, and boundedness give

$$c\|e\|_{1,\Omega}^2 \leq a(e, e) = a(e, y - v_h) \leq C\|e\|_{1,\Omega}\|y - v_h\|_{1,\Omega}.$$

If $e \neq 0$, division by $\|e\|_{1,\Omega}$ gives $\|e\|_{1,\Omega} \leq (C/c)\|y - v_h\|_{1,\Omega}$; the same inequality is immediate if $e = 0$. Since v_h was arbitrary, taking the infimum over V_h^0 proves

$$\|y - y_h\|_{1,\Omega} \leq \frac{C}{c} \inf_{v_h \in V_h^0} \|y - v_h\|_{1,\Omega}.$$